

MUBARAK ADESOLA ADEDEJI

Data Analyst | AI Developer | Public Health Data | Field Research | M&E

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PROFESSIONAL SUMMARY

Data Analyst and AI Developer with 3+ years of end-to-end experience building data systems, leading field research, and deploying AI applications across public health, government, and commercial environments. Currently serving as Junior Data Analyst at Natview Foundation for Technology Innovation, supporting data management, field supervision, and M&E operations across multiple project sites in Kaduna State. Delivered a zero-error longitudinal dataset across 80+ PHC facilities in 14 LGAs under Longitudinal Study 2.0. Built a SQL data warehouse and Power BI dashboard that fed into the state-wide HEFA dashboard used by the entire Kaduna State health system. Developed and deployed 4 live AI and ML applications including an LLM clinical data validator achieving 97.7% accuracy with zero hallucinations. Fluent in Python, SQL, R, Power BI, and ODK. Open to remote and international opportunities.

PROFESSIONAL EXPERIENCE

Natview Foundation for Technology Innovation

Feb 2026 – Present

Junior Data Analyst

Jun 2026 – Present

- **Key Responsibility:** Supporting data management objectives across assigned projects by designing and maintaining data workflows, digitising data collection tools, and automating reporting processes from both primary and secondary sources.
- Collaborating with project teams to identify data sources, define collection processes, and optimise data preparation exercises to ensure datasets are accurate, complete, and analysis-ready.
- Applying statistical and analytical tools in Python and R to interpret datasets, identify trends and patterns, and answer diagnostic and predictive analytics questions for programme decision-making.
- Developing SOPs for data governance and creating documentation to support knowledge management and knowledge transfer to key stakeholders across programmes.
- Supporting the production of project presentations, briefs, and research findings for dissemination to internal teams and external partners.

Field Data Enumerator | Longitudinal Study 2.0

Feb 2026 – Present

- **Key Achievement:** Delivered a zero-error longitudinal health dataset across 80+ PHC facilities in 14 LGAs in Kaduna State by designing and deploying a structured data collection instrument with built-in validation logic and cross-field checks.
- Personally visited and assessed 30+ health facilities across Kaduna State LGAs, verifying vaccine availability, cold chain records, essential medicine stock levels, utilisation history, supply chain documentation, staff attendance, and cadre records against source documents on every visit.
- Maintained field data quality across a 14-LGA operation by supervising data collectors, resolving discrepancies in real time, and ensuring all facility records were complete and compliant before submission.
- Supported routine health data reporting and cross-verification of facility-level records against source documents using DHIS2, contributing to a dataset now used in state-level immunisation planning.

Natview Technology

Mar 2026 – Present

ICT Instructor and School Supervisor | PDS-FAST Project | World Bank Partnership

Mar 2026 – Present

- **Key Achievement:** Designed and delivered a 5-day Train-the-Trainer ICT programme for 15+ teachers across 6 schools in Kaduna State, covering LMS deployment, ICT-integrated lesson planning, online assessment design, and digital classroom management.
- Assessed teacher readiness through structured microteaching sessions, evaluating lesson plan quality, LMS proficiency, tool justification, learner engagement, and formative assessment practice against the PDS-FAST digital competency framework.
- Adapted training delivery in real time to suit low-connectivity school environments, including offline microteaching workarounds, ensuring all 15+ teachers achieved minimum competency standards regardless of infrastructure constraints.

- Supervised school-level implementation across all training sites, monitoring teacher progress and supporting the development of practical ICT outputs including lesson plans, LMS course sections, quizzes, assignments, and teaching materials.
- Contributed to a World Bank-partnered initiative aligned with UNESCO competency standards and the National Digital Literacy Framework (NDLF) to close digital skills gaps in Nigerian secondary schools.
- Reported directly to the Project Manager on training outcomes, teacher performance data, and ICDL pathway candidate selection across all participating schools.

Click-On Kaduna *Data Science Fellow*

Oct 2025 – Apr 2026

- **Key Achievement:** Cut analyst query time by ~40% across a longitudinal health dataset spanning 14 LGAs by designing and building a Bronze-Silver-Gold SQL data warehouse with repeatable ETL workflows and structured validation at each layer.
- Contributed directly to the state-wide HEFA dashboard used by the entire Kaduna State health system by rebuilding the PHC facility readiness Power BI dashboard, redesigning the data model and adding DAX trend measures across 500+ facility records.
- Designed a 60-variable ODK survey instrument for a maternal health prevalence study, structuring each field with range checks and cross-field validation rules to ensure zero entry errors across the full collection period.
- Surfaced PHC readiness gaps and coverage patterns that fed into Kaduna State planning recommendations by running EDA and statistical analysis in Python and R on longitudinal facility data.
- Supported routine health data reporting and cross-verification of facility records against source documents using DHIS2.
- Mentored 20+ data science students in Python, Excel, and analytical thinking as a student tutor, guiding capstone projects built on real datasets.

Bilsun Textile LTD *Lead Data Analyst*

Sep 2023 – Mar 2026

- **Key Achievement:** Reduced overstock by 15% (confirmed by management) and cut reporting time by ~30% by building a Power BI and Excel reporting suite that replaced manual spreadsheet workflows across 5 product lines.
- Enabled automated stock reorder decisions by modelling demand patterns across 1,000+ monthly transactions using trend decomposition across 18 months of sales data.
- Eliminated ongoing analyst dependency by developing pivot-table dashboards with automated Excel formulas, giving management real-time revenue visibility without manual input.

Lagos State Ministry of Innovation, Science & Technology *Data & Policy Analyst (SIWES Trainee)*

Apr 2024 – Oct 2024

- **Key Achievement:** Accelerated evidence-based decision-making by ministry officials by synthesising 20+ digital governance policy documents into concise, decision-ready analytical briefs.
- Improved reporting clarity and reduced duplication across technology policy teams by mapping and documenting public-sector data workflows into a structured reference framework.

PROJECTS

Clinical NLP Validator *Python | Groq API | Streamlit | [Live App](#) | [GitHub](#)*

- **Problem:** LLMs extracting structured data from clinical notes produce errors that enter pipelines silently with no systematic validation layer.
- **Approach:** Built a validation framework scoring 11 fields per note against ground truth across 4 outcome categories (correct, wrong, missing, hallucinated) across 20 clinical notes. Deployed as a live Streamlit app.
- **Outcome:** Achieved 97.7% mean accuracy and zero hallucinations across 220 field evaluations. 8 of 11 fields cleared for production without human review.

Diabetes Risk Predictor *Python | Scikit-learn | Streamlit | [Live App](#) | [GitHub](#)*

- **Problem:** Standard ML classifiers using default thresholds miss high-risk patients, reducing the clinical utility of prediction models.
- **Approach:** Trained and compared 4 models using 5-fold stratified cross-validation. Engineered 4 domain-informed features. Optimised classification threshold from 0.50 to 0.26.

- **Outcome:** Reduced missed diabetic patients from 27 to 6 out of 54, a 38.9 percentage point improvement in clinical recall. Deployed as a live interactive app.

Nftify: RAG Question Answering System *Python* | *ChromaDB* | *Sentence Transformers* | *Gemini* | *Streamlit* | [Live App](#) | [GitHub](#)

- **Problem:** Generic LLMs hallucinate or fail entirely on specialised queries about NFTI and Click-On Kaduna.
- **Approach:** Built a full RAG pipeline: web scraping with Trafilatura, recursive text chunking, all-MiniLM-L6-v2 embeddings, ChromaDB semantic retrieval, and Gemini 2.5 Flash generation.
- **Outcome:** Deployed a modular AI assistant with source-backed answers and zero hallucination on domain-specific queries.

Healthcare A/B Testing Framework *Python* | *SciPy* | *Statsmodels* | [GitHub](#)

- **Problem:** Simulate and validate a randomised controlled trial evaluating a lifestyle intervention on diabetes risk indicators across 768 patients.
- **Approach:** Stratified random split, power analysis confirming 80% power at alpha 0.05, 3 statistical tests per metric, 95% confidence intervals, and Cohen's d effect sizes throughout.
- **Outcome:** Validated correct randomisation with zero significant differences. Produced a structured clinical recommendation with documented study limitations.

TECHNICAL SKILLS

AI & ML	RAG pipeline design ChromaDB Sentence Transformers LLM API integration (Gemini, Groq) Scikit-learn LLM output validation Streamlit deployment
Programming	Python (pandas, NumPy, Matplotlib, Seaborn, SciPy, Statsmodels) R (ggplot2, tidyR, dplyr) SQL
BI & Visualisation	Power BI (DAX, data modelling, drill-through, trend measures) Excel (pivot tables, advanced formulas, dynamic dashboards)
Data Engineering	ETL pipeline design SQL warehouse architecture (Bronze/Silver/Gold) Data cleaning and validation at scale
Field & M&E	ODK Collect ODK instrument design DHIS2 Facility register verification Immunisation data monitoring Supply chain data Data quality assurance Field supervision
Training & Facilitation	LMS deployment ICT-integrated lesson planning Train-the-Trainer delivery Digital competency assessment SOP development Knowledge management documentation
Statistical Methods	A/B testing Experimental design Hypothesis testing Power analysis Effect size Confidence intervals EDA Feature engineering
Other Tools	Git Jupyter Kaggle HTML/CSS Microsoft Office Suite Canva

EDUCATION

Air Force Institute of Technology (AFIT), Kaduna

Jul 2022 – Jun 2026

B.Sc. Mathematics | Expected June 2026

Relevant Coursework: Linear Algebra, Probability Theory, Mathematical Statistics, Mathematical Modelling, Numerical Methods, Real Analysis

CERTIFICATIONS

- **Python Data Associate** | DataCamp
- **SQL Associate** | DataCamp
- **AI Fundamentals** | DataCamp
- **Understanding Data Science** | DataCamp
- **Green Digital Certificate Program** | INCO Academy
- **Data Science Fellowship** | Click-On Kaduna

LEADERSHIP & COMMUNITY

TheSoftwareChasers *Co-Founder & Project Manager*

Jun 2022 – Present

- Co-founded a tech startup building Sell Harvest (agro-e-commerce for smallholder farmers) and SkillHive365 (online training platform), leading product development and team coordination from concept to deployment.

SkillHive *Program Manager*

Oct 2021 – Nov 2024

- Managed programme delivery across 100+ learners over 3 years, designing performance tracking dashboards and producing cohort evaluation reports that drove scheduling and resource improvements across 3 programme tracks.

Mathematics Students Association, AFIT *Social Director*

Jul 2023 – May 2025

- Organised academic and community events for a 100+ member student body across the mathematics department.

LANGUAGES

Yoruba | Native English | Fluent (C1/C2) Hausa | Basic Conversational